

Smart Waste Monitoring System

Analytics & Performance Report

Report Generated: 2026-08-26 19:58:39
Generated By: Smart Waste Monitoring System

System Overview

This report presents analytics generated from the smart waste monitoring system. The system analyses bin fill levels, identifies collection priorities using a rule-based DSS, and logs collection events separately from sensor telemetry.

Metric	Value
Total Telemetry Records	1330
Average Fill Level	47.7%
Critical Records (≥80%)	201
Highest Risk Bin	BIN-001
Highest Fill Level	94%
Total Collection Events	99
Most Collected Bin	BIN-002
Avg Fill at Collection	97.8%

Historical Trend Analysis

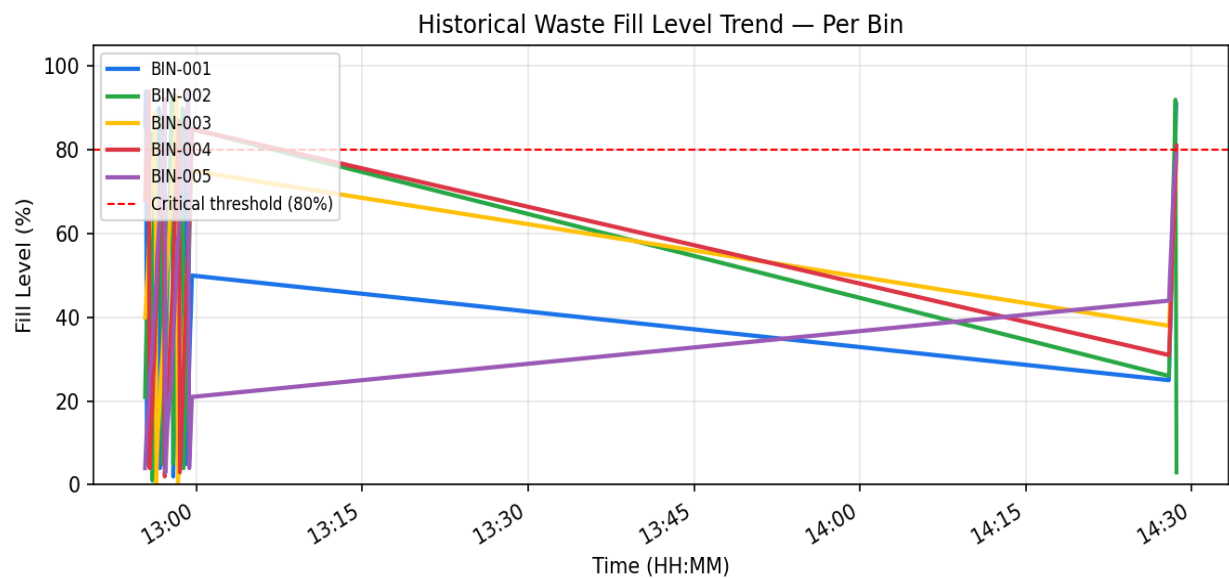


Figure 1. Historical waste fill level trend across recorded sensor readings.

Risk Distribution Analysis

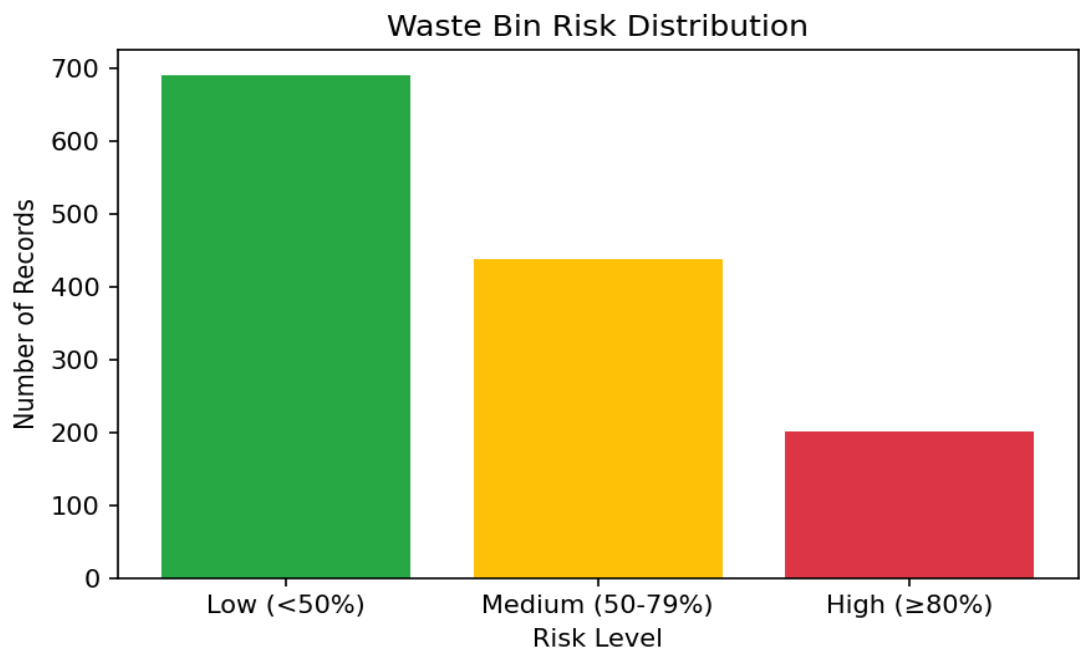


Figure 2. Distribution of low, medium, and high-risk waste records.

Collection Events Summary

The following table summarises bin collection events recorded in the *collection_events* table — stored separately from sensor telemetry to enable clean queryability of actual emptying events.

Bin ID	Location	Fill at Collection (%)	Collected At
BIN-002	Pettah Market	100%	2026-08-26 14:28:36
BIN-005	Wellawatte	100%	2026-08-26 12:59:22
BIN-001	Colombo Fort	96%	2026-08-26 12:59:06
BIN-002	Pettah Market	97%	2026-08-26 12:58:51
BIN-004	Bambalapitiya	95%	2026-08-26 12:58:31
BIN-003	Slave Island	98%	2026-08-26 12:58:20
BIN-001	Colombo Fort	97%	2026-08-26 12:57:55
BIN-002	Pettah Market	100%	2026-08-26 12:57:55
BIN-005	Wellawatte	97%	2026-08-26 12:57:14
BIN-004	Bambalapitiya	95%	2026-08-26 12:57:09

System Assessment

The analytics module successfully evaluates historical waste accumulation trends and identifies high-priority collection requirements. The rule-based DSS applies configurable fill-level thresholds to generate actionable recommendations. Collection events are stored in a dedicated table, enabling independent analysis of collection frequency and bin-level performance.